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Episode 8 · September 18, 2026 · 2 papers

Leader-Follower Formation Control with Prescribed Convergence Rates under Bearing Persistence of Excitation

[SKIM](#)

Tarek Bouazza et al.

They replace the fixed gain in bearing-only leader-follower formation control with a time-varying matrix gain built from each follower's own bearing measurements, so you can dial in the exponential convergence rate instead of inheriting whatever rate the formation's excitation level happens to give you.

PROBLEM In bearing-only formation control, a follower can only recover the missing range information if the bearing directions vary enough over time, the persistence of excitation condition. With a fixed gain, the guaranteed convergence rate is tied to the PE bound, so a formation with weak excitation converges slowly and you cannot fix it by tuning. The gap was decoupling rate from excitation quality.

RESULTS The abstract reports no numbers at all. There is no convergence rate figure, no formation size, no noise level, no quantitative baseline gap. The only stated evidence is simulations comparing convergence properties against fixed-gain designs, in an eight-page conference paper with three figures.

METHOD Each follower runs a matrix gain that is itself a dynamical system driven by the bearing vectors it measures, roughly an adaptive gain that accumulates the directions seen so far and inverts that accumulation, in the spirit of least-squares or Riccati-style adaptive observers. The authors prove the gain stays well posed and uniformly bounded whenever the formation is bearing-persistently exciting, and then show exponential convergence at a rate set by a single tunable design parameter rather than by the PE constant. The design is then carried over from single-integrator to double-integrator follower dynamics.

CAVEAT The whole guarantee rests on bearing-persistent excitation, and the paper as described never says what happens when that condition is marginal, intermittent, or violated, which is exactly the regime where a rate-decoupling claim would earn its keep. Inverting an accumulated bearing matrix is numerically delicate near the excitation boundary, and uniform boundedness of the gain is a much weaker promise than good conditioning. Add that measurements are assumed noise-free bearings and velocities, evaluation is simulation only with three figures, and there is no comparison against adaptive or range-estimating alternatives, only against fixed-gain controllers the method is designed to beat.

WHY IT MATTERS This is peripheral to your work, and the distance is worth naming precisely. Bearing-only formation control assumes vehicles sense each other and communicate almost nothing, which is the opposite of the premise you are building on, namely that vehicles exchange intentions and proposed trajectories and negotiate. There is no maneuver protocol here, no proposal or acceptance, no conflict resolution, no packet loss model. The one transferable idea is the structure of the argument, that convergence speed can be decoupled from how informative the incoming signal is, which has a loose analogy to platoon string stability degrading with communication quality. That analogy is suggestive, not a result you can cite.

arXiv:2609.19106v1 PDF [incremental](#)

Social Laws for Multi-agent Coordination in Stochastic Environments

SKIM

Rolando Fernandez et al.

They take social laws, the old idea that you constrain what agents are allowed to do so they stop getting in each other's way, and extend it from deterministic goal-reaching to stochastic reward settings, with a way to check how much utility a law guarantees you.

PROBLEM Social law theory was built for deterministic, goal-based worlds where the question is whether an agent can still reach its goal under the restriction. That leaves open the case where outcomes are probabilistic and agents care about accumulated reward, not a binary goal, so there was no way to say what a law actually costs you.

RESULTS The abstract gives no numbers at all. It says only that empirical evaluation is on toy environments and that the results illustrate the potential of the framework. There are no baselines, no task names, no utility figures.

METHOD They define alpha-robustness: the guaranteed utility an agent keeps when it runs its own single-agent optimal policy under the assumption that everyone else obeys the law. Verification is done by reducing the question to solving a series of Markov decision processes, so checking a law becomes a planning problem rather than a bespoke proof. The key move is that each agent plans as if alone, and the law, not inference about others, is what carries the coordination.

CAVEAT The evaluation is toy environments with no reported quantities, which is thin even for a workshop paper, but the sharper issue is the assumption baked into alpha-robustness itself: it assumes all other agents obey the law. That is exactly the case that does not need much verification. There is no treatment of what happens when a fraction of agents defect or simply cannot comply, and the MDP reduction is likely to scale badly as the number of agents and the joint state space grow.

WHY IT MATTERS This is adjacent rather than central. The conceptual link is real: a social law is the limit case of your field, coordination by prior convention instead of by message exchange, and alpha-robustness is a clean way to ask what utility a vehicle sacrifices by following a rule like always yield to the vehicle on the right. But it does not model communication, so it says nothing about message loss, delay, or partial penetration, and those are the failure modes you care about. The honest read is that it offers one vocabulary item, a quantified price of obedience, not a method you would adopt.

arXiv:2609.18929v1 PDF incremental